

Machine Learning Workflow Notes Classification and Regression

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1 Introduction

In Machine Learning, both **classification** and **regression** problems follow a similar workflow. The main difference lies in the type of output and the evaluation metrics used.

- **Classification:** Predicts discrete class labels (e.g., spam or not spam).
- **Regression:** Predicts continuous values (e.g., house price prediction).

The following steps describe the general process used to train and evaluate machine learning models.

2 Step 1: Import Required Libraries

First, import the necessary libraries required for data processing, model training, and evaluation.

Common libraries include:

- NumPy
- Pandas
- Scikit-learn

Example:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
```

3 Step 2: Load the Dataset

Load the dataset from a file or built-in dataset.

- **X:** Input features (independent variables)
- **y:** Target variable (dependent variable)

Example:

```
X = data.features
y = data.target
```

4 Step 3: Data Preprocessing

Data preprocessing prepares the dataset before training the model.

Common preprocessing tasks include:

- Handling missing values
- Encoding categorical variables
- Feature scaling
- Feature selection

Example (Feature Scaling):

```
from sklearn.preprocessing import StandardScaler
```

5 Step 4: Split the Dataset

The dataset is divided into two parts:

- **Training Set:** Used to train the model
- **Testing Set:** Used to evaluate the model

Example:

```
X_train, X_test, y_train, y_test =  
train_test_split(X, y, test_size=0.2)
```

Typically:

- 80
- 20

6 Step 5: Feature Scaling

Feature scaling ensures that all input features are in a similar range.

This step is important for algorithms such as:

- Neural Networks
- Support Vector Machines (SVM)
- K-Nearest Neighbors (KNN)

Example:

```
scaler = StandardScaler()  
  
X_train = scaler.fit_transform(X_train)  
X_test = scaler.transform(X_test)
```

Important:

- Fit the scaler only on training data
- Apply the transformation to test data
- This avoids data leakage

7 Step 6: Select and Define the Model

Choose an appropriate machine learning model depending on the problem.

Classification Models

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Neural Networks

Regression Models

- Linear Regression
- Polynomial Regression
- Ridge / Lasso Regression
- Support Vector Regression (SVR)
- Neural Network Regression

8 Step 7: Train the Model

Train the model using the training dataset.

Example:

```
model.fit(X_train, y_train)
```

During training, the model learns patterns and relationships from the data.

9 Step 8: Make Predictions

After training, the model is used to make predictions on the test dataset.

Example:

```
y_pred = model.predict(X_test)
```

10 Step 9: Evaluate the Model

Model evaluation depends on whether the task is classification or regression.

Classification Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Example:

```
accuracy_score(y_test, y_pred)
```

Regression Evaluation Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2 score)

Example:

```
mean_squared_error(y_test, y_pred)
```

11 Summary of Workflow

1. Import required libraries
2. Load the dataset
3. Perform data preprocessing
4. Split the dataset into training and testing sets
5. Apply feature scaling if required
6. Select and define the machine learning model
7. Train the model using training data
8. Make predictions on the test dataset
9. Evaluate the model using appropriate metrics

12 Conclusion

Both classification and regression follow the same machine learning workflow. The main difference lies in the type of prediction and evaluation metrics used. Following these steps ensures that the model is trained properly and evaluated correctly.

13 Note

The code given in this note is for the understanding purpose only.